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ICEMAKER AND ICE STORAGE SYSTEM FOR A REFRIGERATOR APPLIANCE

Abstract

A refrigerator appliance includes a cabinet defining a chilled chamber, a door rotatably mounted to the cabinet and rotatable between a closed position enclosing the chilled chamber and an open position providing access to the chilled chamber, an icemaker mounted to the door, and an ice storage system for storing ice produced by the icemaker. The ice storage system includes a door-mounted ice bucket positioned below the icemaker for storing the ice formed by the icemaker; an ice storage drawer positioned within the chilled chamber, and a damper mounted at a bottom end of the door-mounted ice bucket, the damper being movable between a closed position to retain the ice within the door-mounted ice bucket and an open position to direct the ice into the ice storage drawer.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present subject matter relates generally to refrigerator appliances, and more particularly to storing ice made in a door-mounted icemaker.

BACKGROUND OF THE INVENTION

[0002] Refrigerator appliances generally include a cabinet that defines one or more chilled chambers for receipt of food articles for storage. Typically, one or more doors are rotatably hinged to the cabinet to permit selective access to food items stored in the chilled chamber. Further, refrigerator appliances commonly include ice making assemblies mounted within an icebox on one of the doors or in a freezer compartment. The ice is stored in a storage bin and is accessible from within the freezer chamber or may be discharged through a dispenser recess defined on a front of the refrigerator door.

[0003] It may be desirable to place craft icemakers on the freezer door for forming craft ice cubes (e.g., such as balls of ice greater than 2 inches in diameter), which are typically large cubes made by a conventional twist tray icemaker. Conventional craft icemakers may be positioned below the primary icemaker on the refrigerator door. However, in many cases, accessibility of the ice in the storage bin with the door open may be hampered by storage bin being fixed to the icemaker. For dual ice making systems on the door, the second ice maker does not have an ice dispensing recess outside of the door, and customers may have difficulty accessing the bottom ice bucket.

Furthermore, the ice storage capacity of the storage bin may be insufficient during high demand periods, e.g., when a consumer is preparing for a large gathering.

[0004] Accordingly, a refrigerator appliance with features for improved ice making would be desirable. More particularly, a refrigerator appliance with a door-mounted craft icemaker that has a high capacity and an easily accessible ice storage bin would be particularly beneficial.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, may be apparent from the description, or may be learned through practice of the invention.

[0006] In one exemplary embodiment, a refrigerator appliance defining a vertical direction, a lateral direction, and a transverse direction is provided, including a cabinet defining a chilled chamber, a door rotatably mounted to the cabinet and rotatable between a closed position enclosing the chilled chamber and an open position providing access to the chilled chamber, an icemaker mounted to the door, and an ice storage system for storing ice produced by the icemaker. The ice storage system includes a door-mounted ice bucket positioned below the icemaker for storing the ice formed by the icemaker, an ice storage drawer positioned within the chilled chamber, and a damper mounted at a bottom end of the door-mounted ice bucket, the damper being movable between a closed position to retain the ice within the door-mounted ice bucket and an open position to direct the ice into the ice storage drawer.

[0007] In another exemplary embodiment, an ice storage system for storing ice produced by an icemaker mounted to a door of a refrigerator appliance is provided. The ice storage system includes a door-mounted ice bucket positioned below the icemaker for storing the ice formed by the icemaker, an ice storage drawer positioned within a chilled chamber of the refrigerator appliance, and a damper mounted at a bottom end of the door-mounted ice bucket, the damper being movable between a closed position to retain the ice within the door-mounted ice bucket and an open position to direct the ice into the ice storage drawer.

[0008] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying

drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

[0010] FIG. 1 provides a perspective view of a refrigerator appliance according to an example embodiment of the present subject matter.

[0011] FIG. 2 provides a front view of the example refrigerator appliance of FIG. 1, with the doors of the fresh food chamber and freezer chamber shown in an open position.

[0012] FIG. 3 provides a perspective cross-sectional view of an icebox and ice making assembly for use with the example refrigerator appliance of FIG. 1 according to an example embodiment of the present subject matter.

[0013] FIG. 4 provides a perspective view of an icemaker of the example refrigerator appliance of FIG. 1 according to an example embodiment of the present subject matter.

[0014] FIG. 5 provides a perspective view of an ice storage drawer of the example refrigerator appliance of FIG. 1 according to an example embodiment of the present subject matter.

[0015] FIG. 6 provides a perspective view of a door of the example refrigerator appliance of FIG. 1 including an icebox and icemaker according to an example embodiment of the present subject matter.

[0016] FIG. 7 provides a side view of an example door-mounted ice bucket that may be used with the example refrigerator appliance of FIG. 1 according to an example embodiment of the present subject matter.

[0017] FIG. 8 provides a side view of an example door-mounted ice bucket that may be used with the example refrigerator appliance of FIG. 1 according to an example embodiment of the present subject matter.

[0018] FIG. 9 provides a perspective view of a door of the example refrigerator appliance of FIG. 1 including an icebox and icemaker according to an example embodiment of the present subject matter.

[0019] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0020] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0021] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). The term “at least one of” in the

context of, e.g., “at least one of A, B, and C” refers to only A, only B, only C, or any combination of A, B, and C. In addition, here and throughout the specification and claims, range limitations may be combined and/or interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0022] Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “generally,” “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 10 percent margin, i.e., including values within ten percent greater or less than the stated value. In this regard, for example, when used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction, e.g., “generally vertical” includes forming an angle of up to ten degrees in any direction, e.g., clockwise or counterclockwise, with the vertical direction V.

[0023] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” In addition, references to “an embodiment” or “one embodiment” does not necessarily refer to the same embodiment, although it may. Any implementation described herein as “exemplary” or “an embodiment” is not necessarily to be construed as preferred or advantageous over other implementations. Moreover, each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0024] FIG. 1 provides a perspective view of a refrigerator appliance **100** according to an exemplary embodiment of the present subject matter. Refrigerator appliance **100** includes a cabinet or housing **102** that extends between a top **104** and a bottom **106** along a vertical direction V, between a first side **108** and a second side **110** along a lateral direction L, and between a front side **112** and a rear side **114** along a transverse direction T. Each of the vertical direction V, lateral direction L, and transverse direction T are mutually perpendicular to one another.

[0025] Housing **102** defines chilled chambers for receipt of food items for storage. In particular, housing **102** defines fresh food chamber **122** positioned at or adjacent second side **110** of housing **102** and a freezer chamber **124** arranged at or adjacent first side **108** of housing **102**. As such, refrigerator appliance **100** is generally referred to as a side-by-side refrigerator. It is recognized, however, that the benefits of the present disclosure apply to other types and styles of refrigerator appliances such as, e.g., a top mount refrigerator appliance, a bottom mount refrigerator appliance, or a single door refrigerator appliance. Consequently, the description set forth herein is for illustrative purposes only and is not intended to be limiting in any aspect to any particular refrigerator chamber configuration.

[0026] A refrigerator door **128** is rotatably hinged to an edge of housing **102** for selectively accessing fresh food chamber **122**. In addition, a freezer door **130** is rotatably hinged to an edge of housing **102** for selectively accessing freezer chamber **124**. Refrigerator door **128** and freezer door **130** are shown in the closed configuration in FIG. 1. One skilled in the art will appreciate that other chamber and door configurations are possible and within the scope of the present invention.

[0027] FIG. 2 provides a front view of refrigerator appliance **100** shown with refrigerator door **128** and freezer door **130** in the open position. As shown in FIG. 2, various storage components are mounted within fresh food chamber **122** to facilitate storage of food items therein as will be understood by those skilled in the art. In particular, the storage components may include bins **134** and shelves **136**. Each of these storage components are configured for receipt of food items (e.g., beverages and/or solid food items) and may assist with organizing such food items. As illustrated, bins **134** may be mounted on refrigerator door **128** and freezer door **130** or may slide into a receiving space in fresh food chamber **122** or freezer chamber **124**. It should be appreciated that the illustrated storage components are used only for the purpose of explanation and that other storage components may be used and may have different sizes, shapes, and configurations.

[0028] Referring now generally to FIG. 1, a dispensing assembly **140** will be described according to exemplary embodiments of the present subject matter. Dispensing assembly **140** is generally configured for dispensing liquid water and/or ice. Although an exemplary dispensing assembly **140** is illustrated and described herein, it should be appreciated that variations and modifications may be made to dispensing assembly **140** while remaining within the present subject matter.

[0029] Dispensing assembly **140** and its various components may be positioned at least in part within a dispenser recess **142** defined on freezer door **130**. In this regard, dispenser recess **142** is defined on a front side **112** of refrigerator appliance **100** such that a user may operate dispensing assembly **140** without opening freezer door **130**. In addition, dispenser recess **142** is positioned at a predetermined elevation convenient for a user to access ice and enabling the user to access ice without the need to bend-over. In the exemplary embodiment, dispenser recess **142** is positioned at a level that approximates the chest level of a user.

[0030] Dispensing assembly **140** includes an ice dispenser **144** including a discharging outlet **146** for discharging ice from dispensing assembly **140**. An actuating mechanism **148**, shown as a paddle, is mounted below discharging outlet **146** for operating ice or water dispenser **144**. In alternative exemplary embodiments, any suitable actuating mechanism may be used to operate ice dispenser **144**. For example, ice dispenser **144** can include a sensor (such as an ultrasonic sensor) or a button rather than the paddle. Discharging outlet **146** and actuating mechanism **148** are an external part of ice dispenser **144** and are mounted in dispenser recess **142**.

[0031] As shown in FIGS. 2 and 3, inside refrigerator appliance **100**, freezer door **130** may define an icebox **150** housing one or more icemakers and ice storage bins **152** that are configured to form ice. In this regard, for example, icebox **150** may define an ice making chamber **154** for housing ice making assemblies, storage mechanisms, and dispensing mechanisms. According to the illustrated embodiment, icebox **150** may include dispensing assembly **140** and may have a main icemaker. In addition, as described in more detail below, icebox **150** may include an icemaker for forming “craft ice” that is commonly large, clear cubes or spheres of ice for alcoholic or non-alcoholic drinks. For example, a user may access this craft ice by opening freezer door **130** and accessing storage bin **152** directly.

[0032] A control panel **160** is provided for controlling the mode of operation. For example, control panel **160** includes one or more selector inputs **162**, such as knobs, buttons, touchscreen interfaces, etc., such as a water dispensing button and an ice-dispensing button, for selecting a desired mode of operation such as crushed or non-crushed ice. In addition, inputs **162** may be used to specify a fill volume or method of operating dispensing assembly **140**. In this regard, inputs **162** may be in communication with a processing device or controller **164**. Signals generated in controller **164** operate refrigerator appliance **100** and dispensing assembly **140** in response to selector inputs **162**. Additionally, a display **166**, such as an indicator light or a screen, may be provided on control panel **160**. Display **166** may be in communication with controller **164** and may display information in response to signals from controller **164**.

[0033] As used herein, “processing device” or “controller” may refer to one or more microprocessors or semiconductor devices and is not restricted necessarily to a single element. The

processing device can be programmed to operate refrigerator appliance **100** and dispensing assembly **140**. The processing device may include, or be associated with, one or more memory elements (e.g., non-transitory storage media). In some such embodiments, the memory elements include electrically erasable, programmable read only memory (EEPROM). Generally, the memory elements can store information accessible processing device, including instructions that can be executed by processing device. Optionally, the instructions can be software or any set of instructions and/or data that when executed by the processing device, cause the processing device to perform operations.

[0034] According to the illustrated embodiments, refrigerator appliance **100** may include a first icemaker or an upper icemaker **200** (e.g., a main icemaker associated with dispensing assembly **140** for discharging ice through ice dispenser **144**) and a second icemaker or a lower icemaker **202** (e.g., a secondary icemaker such as a craft icemaker). According to example embodiments, each of upper icemaker **200** and lower icemaker **202** are mounted to freezer door **130**, e.g., within icebox **150**. According to example embodiments, each of upper icemaker **200** and lower icemaker **202** may have a dedicated storage bin **152** for storing ice. For example, upper icemaker **200** may have a cover that allows chilled air to flow in from freezer chamber **124** to increase icemaking rate. By contrast, lower icemaker **202** may have a cover to slow the ice freezing rate, e.g., which may facilitate the production of higher quality cubes.

[0035] In addition, according to an example embodiment, upper icemaker **200** may include a dispensing type of bucket and motor that would supply ice to discharging outlet **146** of dispensing assembly **140** (or alternatively could include an ice storage bin with no external dispenser). By contrast, lower icemaker **202** may include an ice bin that has no dispensing system. Refrigerator appliance **100** may further include an ice storage system **204** that is generally configured for storing ice (e.g., as identified by reference numeral **206** in FIG. **6**) that is produced by one or both of upper icemaker **200** and lower icemaker **202**. Although an exemplary construction of ice storage system **204** is described herein, it should be appreciated that variations and modifications may be made to ice storage system **204** while remaining within the scope of the present subject matter.

[0036] Referring now generally to FIGS. **2** through **9**, ice storage system **204** may include a door-mounted ice bucket **210** that is positioned below lower icemaker **202** for storing craft ice **206** formed therein. In this regard, lower icemaker **202** is positioned above the door-mounted ice bucket **210** such that harvested ice **206** may fall from lower icemaker **202** into the door-mounted ice bucket **210**. Notably, as explained briefly above, the volume within door-mounted ice bucket **210** is relatively limited and there are times where a consumer may wish to store a larger volume of craft ice **206**, e.g., before hosting a large party. Accordingly, aspects of the present subject matter are directed to an ice storage system **204** that includes features for expanded ice storage.

[0037] Specifically, as best shown in FIGS. **3** and **5**, ice storage system **204** may further include an ice storage drawer **212** that is positioned within a chilled chamber of refrigerator appliance **100** (e.g., within freezer chamber **124**). As will be explained in more detail below, ice storage drawer **212** may be configured for receipt of excess ice **206** formed by lower icemaker **202** when it is desirable to store larger volumes of ice **206**. As shown, ice storage drawer **212** is not attached to freezer door **130** and thus remains within freezer chamber **124** when freezer door **130** is in the open position. According to the illustrated embodiment, ice storage drawer **212** is fixed within freezer chamber **124**. However, it should be appreciated that according to alternative embodiments, ice storage drawer **212** may be slidably mounted within freezer chamber **124**.

[0038] Ice storage system **204** may further include features for supplying or directing ice **206** from door-mounted ice bucket **210** into ice storage drawer **212**. For example, ice **206** may be transmitted into ice storage drawer **212** when freezer door **130** is in the closed position and a consumer has requested the production of a larger volume of ice. Specifically, according to the illustrated embodiment, a damper **220** may be pivotally mounted over an aperture **222** defined in a bottom wall **224** of door-mounted ice bucket **210**. Damper **220** may generally be movable between a closed

position where ice **206** is retained within door-mounted ice bucket **210** and an open position where ice **206** falls under the force of gravity through aperture **222** and into ice storage drawer **212**. For example, lower icemaker **202** may be programmed to produce ice when damper **220** is in the closed position. After an ice harvest procedure, damper **220** may be selectively activated open and permit formed ice **206** to fall into ice storage drawer **212**.

[0039] As illustrated, ice storage system **204** may further include a receiving ramp **230** that is positioned proximate a front **232** of ice storage drawer **212**. In general, damper **220** may be seated over receiving ramp **230** when damper **220** is in the open position. According to the illustrated embodiment, receiving ramp **230** extends from front **232** of ice storage drawer **212**. However, it should be appreciated that according to alternative embodiments, receiving ramp **230** may extend from any other suitable structure.

[0040] Ice storage system **204** may further include features for facilitating the rotation of damper **220** and the monitoring of ice levels stored within ice storage drawer **212**. For example, referring now specifically to FIG. **4**, ice storage system **204** may include a drive motor **240** that is operably coupled to damper **220** for moving damper **220** between the closed position in the open position. For example, drive motor **240** may be mounted to a bottom wall **224** of door-mounted ice bucket **210** for selectively rotating damper **220**.

[0041] As illustrated, ice storage system **204** may further include a switching mechanism such as a microswitch **242** that is generally configured for detecting when damper **220** reaches the open position. In this regard, microswitch **242** may be physically triggered when damper **220** reaches the position illustrated in FIG. **4**. Notably, if microswitch **242** is not triggered after drive motor **240** has attempted to move damper **220** to the open position, this may be indicative of a full ice storage drawer **212**. Accordingly, monitoring that status of microswitch **242** may provide controller **164** with knowledge of the ice level within ice storage drawer **212**. It should be appreciated that according to alternative embodiments, ice storage system **204** may further include an ice level sensor such as an infrared sensor **244** that is mounted to ice storage drawer **212** to detect ice levels within ice storage drawer **212**.

[0042] Referring now also to FIGS. **6** through **9**, an alternative embodiment of ice storage system **204** will be described. Due to the similarity among embodiments, like reference numerals may be used to refer to the same or similar features. Specifically, according to the illustrated embodiment of FIGS. **6** through **9**, door-mounted ice bucket **210** may be pivotable relative to freezer door **130**. Specifically, door-mounted ice bucket **210** may define an arcuate slot **250** on each side of door-mounted ice bucket **210**. Similarly, the freezer door **130** may define pins **252** that are positioned on opposite sides of door-mounted ice bucket **210**. Pins **252** are received within arcuate slot **250** to permit rotation of door-mounted ice bucket **210** between an upright position (e.g., as shown in FIG. **7**) and a pivoted position (e.g., as shown in FIG. **8**). It should be appreciated that other pivoting mechanisms are possible and within the scope of the present subject matter.

[0043] Referring now specifically to FIG. **9**, according to an alternative embodiment, door-mounted ice bucket **210** may be a multi-part bucket that permits removal of the ice bucket. Specifically, according to the illustrated embodiment, door-mounted ice bucket **210** may include a pivoting bucket holder **260** and a removable storage bucket **262** slidably received within pivoting bucket holder **260**. In this manner, the user may pivot door-mounted ice bucket **210** away from freezer door **130** and may remove storage bucket **262** for easy access to ice **206** stored therein. According to example embodiments, a magnet **264** and a metal plate **266** may be used to magnetically hold the door-mounted ice bucket **210** in the closed position.

[0044] As explained herein, aspects of the present subject matter are generally directed to a refrigerator appliance having two ice makers on a door. For example, the first icemaker is a primary icemaker and the second ice maker may be a craft icemaker and its ice cubes may be stored in the freezer bottom drawer automatically when customers want larger amount of ice. The second icemaker's ice bucket may include a damper beneath the ice bucket and that is normally is closed.

When the damper is in closed condition the second ice maker stores ice cubes in the secondary ice bucket. When customers select to store a larger volume of ice cubes, the damper may open right after harvesting (while door is closed) to dispense the ice in the freezer bottom drawer. The damper may also be used to check whether the ice drawer is full.

[0045] According to example embodiment, a secondary icemaker is mounted to refrigerator inner door panel below the dispenser recess includes an ice bin below the secondary icemaker. The secondary icemaker is located on the freezer door below waist level. The ice bin may be mounted to the slides of the icemaker cover. The ice bin can be pulled forward to access the ice stored inside the bin. A magnet and metal plate magnetically holds the ice bin closed when not in use. According to example, embodiments, the ice bin may be pivotable to improve access to the ice stored therein.

[0046] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A refrigerator appliance defining a vertical direction, a lateral direction, and a transverse direction, comprising: a cabinet defining a chilled chamber; a door rotatably mounted to the cabinet and rotatable between a closed position enclosing the chilled chamber and an open position providing access to the chilled chamber; an icemaker mounted to the door; and an ice storage system for storing ice produced by the icemaker, the ice storage system comprising: a door-mounted ice bucket positioned below the icemaker for storing the ice formed by the icemaker; an ice storage drawer positioned within the chilled chamber; and a damper mounted at a bottom end of the door-mounted ice bucket, the damper being movable between a closed position to retain the ice within the door-mounted ice bucket and an open position to direct the ice into the ice storage drawer.
2. The refrigerator appliance of claim 1, wherein the ice storage drawer is not attached to the door and remains within the chilled chamber when the door is in the open position.
3. The refrigerator appliance of claim 1, wherein the ice storage drawer is slidably mounted within the chilled chamber.
4. The refrigerator appliance of claim 1, wherein the ice storage system further comprises: a receiving ramp positioned proximate a front of the ice storage drawer, wherein the damper is seated over the receiving ramp in the open position.
5. The refrigerator appliance of claim 1, wherein the icemaker produces ice when the damper is in the closed position.
6. The refrigerator appliance of claim 1, wherein the ice storage system further comprises: a drive motor operably coupled to the damper for moving the damper between the closed position and the open position.
7. The refrigerator appliance of claim 1, wherein the ice storage system further comprises: a microswitch for detecting when the damper reaches the open position.
8. The refrigerator appliance of claim 1, wherein the ice storage system further comprises: an infrared sensor mounted to the ice storage drawer to detect when the ice storage drawer is full.
9. The refrigerator appliance of claim 1, wherein the door-mounted ice bucket is pivotable relative to the door.
10. The refrigerator appliance of claim 9, wherein the door-mounted ice bucket defines an arcuate slot and the door defines a pin that is received within the arcuate slot to permit pivoting of the door-

mounted ice bucket.

- 11.** The refrigerator appliance of claim 9, wherein the door-mounted ice bucket comprises a pivoting bucket holder and a removable storage bucket slidably received within the pivoting bucket holder.
 - 12.** The refrigerator appliance of claim 1, wherein the icemaker is a lower icemaker, the refrigerator appliance further comprising an upper icemaker vertically above the lower icemaker, wherein the upper icemaker dispenses ice through a dispenser recess defined on a front of the door.
 - 13.** The refrigerator appliance of claim 1, wherein the refrigerator appliance is a side-by-side refrigerator appliance and the chilled chamber is a freezer chamber.
 - 14.** An ice storage system for storing ice produced by an icemaker mounted to a door of a refrigerator appliance, the ice storage system comprising: a door-mounted ice bucket positioned below the icemaker for storing the ice formed by the icemaker; an ice storage drawer positioned within a chilled chamber of the refrigerator appliance; and a damper mounted at a bottom end of the door-mounted ice bucket, the damper being movable between a closed position to retain the ice within the door-mounted ice bucket and an open position to direct the ice into the ice storage drawer.
 - 15.** The ice storage system of claim 14, wherein the ice storage drawer is not attached to the door and remains within the chilled chamber when the door is in the open position.
 - 16.** The ice storage system of claim 14, wherein the ice storage drawer is slidably mounted within the chilled chamber.
 - 17.** The ice storage system of claim 14, further comprising: a receiving ramp positioned proximate a front of the ice storage drawer, wherein the damper is seated over the receiving ramp in the open position.
 - 18.** The ice storage system of claim 14, further comprising: a drive motor operably coupled to the damper for moving the damper between the closed position and the open position.
 - 19.** The ice storage system of claim 14, further comprising: a microswitch for detecting when the damper reaches the open position.
 - 20.** The ice storage system of claim 14, wherein the door-mounted ice bucket is pivotable relative to the door.
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